



(Independent) Roman Domination Parameterized by Distance to Cluster

Pradeesha Ashok¹, Gautam K. Das², Arti Pandey³, Kaustav Paul^{3(✉)},
and Subhabrata Paul⁴

¹ International Institute of Information Technology Bangalore, Bengaluru, India
pradeesha@iiitb.ac.in

² Department of Mathematics, Indian Institute of Technology Guwahati,
Guwahati, India
gkd@iitg.ac.in

³ Department of Mathematics, Indian Institute of Technology Ropar,
Rupnagar, India
{arti,kaustav.20maz0010}@iitrpr.ac.in

⁴ Department of Mathematics, Indian Institute of Technology Patna,
Daultapur, India
subhabrata@iitp.ac.in

Abstract. Given a graph $G = (V, E)$, a function $f : V \rightarrow \{0, 1, 2\}$ is said to be a *Roman Dominating function* (RDF) if for every $v \in V$ with $f(v) = 0$, there exists a vertex $u \in N(v)$ such that $f(u) = 2$. A Roman Dominating function f is said to be an *Independent Roman Dominating function* (IRDF), if $V_1 \cup V_2$ forms an independent set, where $V_i = \{v \in V \mid f(v) = i\}$, for $i \in \{0, 1, 2\}$. The total weight of f is equal to $\sum_{v \in V} f(v)$, and is denoted as $w(f)$. The *Roman Domination Number* (resp. *Independent Roman Domination Number*) of G , denoted by $\gamma_R(G)$ (resp. $i_R(G)$), is defined as $\min\{w(f) \mid f \text{ is an RDF (resp. IRDF) of } G\}$. For a given graph G , the problem of computing $\gamma_R(G)$ (resp. $i_R(G)$) is defined as the *Roman Domination problem* (resp. *Independent Roman Domination problem*).

In this paper, we examine structural parameterizations of the (Independent) Roman Domination problem. We propose fixed-parameter tractable (FPT) algorithms for the (Independent) Roman Domination problem in graphs that are k vertices away from a cluster graph. These graphs have a set of k vertices whose removal results in a cluster graph. We refer to k as the distance to the cluster graph. Specifically, we prove the following results when parameterized by the deletion distance k to cluster graphs: we can find the Roman Domination Number (and Independent Roman Domination Number) in time $4^k n^{O(1)}$. In terms of lower bounds, we show that the Roman Domination number can not be computed in time $2^{\epsilon k} n^{O(1)}$, for any $0 < \epsilon < 1$ unless a well-known conjecture, SETH fails. In addition, we also show that the Roman Domination problem, parameterized by distance to cluster, does not admit a polynomial kernel unless $\text{NP} \subseteq \text{coNP/poly}$.

Keywords: Roman Domination · Independent Roman Domination · FPT · Distance to Cluster · Kernel

1 Introduction

The concept of Roman Dominating function originated in an paper by Ian Stewart, titled “Defend the Roman Empire!” [12], published in Scientific American. Given a graph, where every vertex corresponds to a distinct geographical region within the historical narrative of the Roman Empire, the characterization of a location as secured or unsecured is delineated by the Roman Dominating function, denoted as f .

Specifically, a vertex v is said to be unsecured if it lacks stationed legions, expressed as $f(v) = 0$. Conversely, a secured location is one where one or two legions are stationed, denoted by $f(v) \in \{1, 2\}$. The strategic methodology for securing an unsecured area involves the deployment of a legion from a neighboring location. In the fourth century A.D., Emperor Constantine the Great enacted an edict precluding the transfer of a legion from a fortified position to an unfortified one if such an action would leave the latter unsecured. Therefore, it is necessary to first have two legions at a given location ($f(v) = 2$) before sending one legion to a neighbouring location. Considering the substantial costs associated with legion deployment in specific areas, the Emperor aimed to strategically minimize the number of legions required to safeguard the Roman Empire.

Given a graph $G = (V, E)$, a *Roman Dominating function* (RDF) is defined as a function $f : V \rightarrow \{0, 1, 2\}$, where every vertex v , for which $f(v) = 0$ must be adjacent to at least one vertex u with $f(u) = 2$. The weight of an RDF is defined as $w(f) = \sum_{v \in V} f(v)$. The *Roman Domination Number* is defined as $\gamma_R(G) = \min\{w(f) \mid f \text{ is an RDF of } G\}$. While the context is clear, if $f(v) = i$ for some RDF f , then we say that v has label i .

Given a graph $G = (V, E)$, a set $S \subseteq V$ is defined as *independent set* if any two vertices of S are non-adjacent. A function f is referred to as an *Independent Roman Dominating function* (IRDF) if f is an RDF and $V_1 \cup V_2$ is an independent set. The *Independent Roman Domination Number* is defined as $i_R(G) = \min\{w(f) \mid f \text{ is an IRDF of } G\}$. An IRDF f of G with $w(f) = i_R(G)$ is denoted as an $i_R(G)$ -function of G . Given a graph $G = (V, E)$, the problem of computing $i_R(G)$ is known as *Independent Roman Domination problem*.

One of the objectives of parameterized complexity is to identify parameters that render NP-hard problems fixed-parameter tractable (FPT). This is of practical significance because there are often small parameters, aside from solution size, that capture important practical inputs. Hence, it only makes sense to explore problems under a multitude of parameters. There has recently been a lot of research in this area. A key research direction involves identifying a parameter as small as possible, under which a problem becomes fixed-parameter tractable or admits a polynomial-sized kernel. Structural parameterization involves a parameter that is a function of the input structure rather than the standard solution size. A continuing trend in structural parameterization is to study problems

parameterized by the deletion distance to various graph classes where the problem is efficiently solvable.

Our parameter of interest is the ‘distance’ of the graph from a natural class of graphs. Here, ‘distance’ refers to the number of vertices that must be deleted from the graph to belong to the specified class. This paper focuses on one such special class of graphs: cluster graphs, where each connected component of the graph is a clique. Note that both Roman Domination and Independent Roman Domination problems can be easily solved in cluster graphs. Given a graph $G = (V, E)$, the minimum number of vertices that need to be deleted from the graph so that the remaining graph becomes a cluster graph is called *distance to cluster* of G (denoted by CVD size of G).

2 Preliminaries

2.1 Graph Theoretic Notations

This paper only considers simple, undirected, finite and nontrivial graphs. Let $G = (V, E)$ be a graph. n and m will be used to denote the cardinalities of V and E , respectively. $N(v)$ stands for the set of neighbors of a vertex v in V and $N[v] = N(v) \cup \{v\}$. For a set $S \subseteq V$, we define $N(S) = \cup_{v \in S} N(v)$. The number of neighbors of a vertex $v \in V$ defines its *degree*, which is represented by the symbol $\deg(v)$. The maximum degree of the graph will be denoted by Δ . For a set $U \subseteq V$, the notation $\deg_U(v)$ is used to represent the number of neighbors that a vertex v has within the subset U . Additionally, we use $N_U(v)$ to refer to the set of neighbors of vertex v within U . Given a set $S \subseteq V$, $G \setminus S$ is defined as the graph induced on $V \setminus S$, that is $G[V \setminus S]$.

A vertex of degree one is known as a *pendant vertex*. A set $S \subseteq V$ is said to be a *dominating set* if every vertex of $V \setminus S$ is adjacent to some vertex of S . A graph G is said to be a *complete graph* if any two vertices of G are adjacent. A set $S \subseteq V$ is said to be a clique if the subgraph of G induced on S is a complete graph. A graph is said to be a *cluster graph* if every component of the graph is a clique. For every positive integer n , $[n]$ denotes the set $\{1, 2, \dots, n\}$.

Given a graph $G = (V, E)$ and a function $f : V \rightarrow \{0, 1, 2\}$, $f_H : V(H) \rightarrow \{0, 1, 2\}$ is defined to be the function f restricted on H , where H is an induced subgraph of G .

For a graph $G = (V, E)$ and a function $f : V \rightarrow \{0, 1, 2\}$; we define $V_i = \{v \in V \mid f(v) = i\}$ for $i \in \{0, 1, 2\}$. The partition (V_0, V_1, V_2) is said to be *ordered partition* of V induced by f . Note that the function $f : V \rightarrow \{0, 1, 2\}$ and the ordered partition (V_0, V_1, V_2) of V have a one-to-one correspondence. So, when the context is clear, we write $f = (V_0, V_1, V_2)$. Given an RDF $f = (V_0, V_1, V_2)$, (V_1, V_2) is said to be *Roman Dominating pair* corresponding to f . When the context is clear, we write Roman Dominating pair (omitting the notion of f).

2.2 Problem Definitions

Before presenting our results, we formalize the problems considered in the paper as follows:

SET-COVER

Input: An universe U and a collection of subsets of U , $F = \{S_1, \dots, S_m\}$ and a non-negative integer k .

Question: Does there exists k sets $S_{i_1}, \dots, S_{i_k}$ in F , such that $\bigcup_{j=1}^k S_{i_j} = U$?

RD-CVD

Input: A graph $G = (V, E)$, a cluster vertex deletion set S and a non-negative integer ℓ .

Parameter: $|S| = k$.

Question: Does there exists an RDF f on G , with weight at most ℓ ?

RD-VC

Input: A graph $G = (V, E)$, a vertex cover S and a non-negative integer ℓ .

Parameter: $|S| = k$.

Question: Does there exists an RDF f on G , with weight at most ℓ ?

IRD-CVD

Input: A graph $G = (V, E)$, a cluster vertex deletion set S and a non-negative integer ℓ .

Parameter: $|S| = k$.

Question: Does there exists an IRDF f on G , with weight at most ℓ ?

d -HITTING SET

Input: An universe U and a collection of subsets of U , $F = \{S_1, \dots, S_m\}$ such that $|S_i| \leq d$, for all $i \in [m]$ and a non-negative integer k .

Question: Does there exists a subset $U' \subseteq U$, such that $|U'| \leq k$ and $U' \cap S_i \neq \emptyset$, for every $i \in [m]$?

2.3 Parameterized Complexity Notations and Definitions

Conjecture 1. [9] (Strong Exponential Time Hypothesis (SETH)) There is no $\epsilon > 0$ such that $\forall q \geq 3$, q -CNFSAT can be solved in $(2 - \epsilon)^n n^{\mathcal{O}(1)}$ time where n is the number of variables in input formula.

Theorem 1. [7] *The SET-COVER problem can be solved in $2^n(m+n)^{\mathcal{O}(1)}$ time where n is the size of the universe and m is the size of the family of subsets of the universe.*

Due to space constraints, the other definitions and theorems are omitted from the manuscript.

2.4 Related Works

From the parameterized complexity point of view, it is surprising that there does not exist much literature on the Roman Domination problem (except [6, 10]), while the classical dominating set problem is very well studied. Some related literature about the parameterized complexity of the domination problem can be found in [1, 5, 11]. One recent work about the domination problem parameterized by several structural parameters like distance to cluster and distance to split can be found in [8]. The techniques we designed in this paper are adaptations of the technique used in [8], with appropriate modifications to fit our problem.

In [6], Fernau proved that the Roman Domination parameterized by the solution size is $W[2]$ -hard in general graphs, but FPT for planar graphs. He also showed that the same problem parameterized by treewidth is FPT. In [10], Mohannapriya et al. showed that a more generalized problem, that is k -Roman Domination problem parameterized by solution size is $W[1]$ -hard, even for split graphs. To the best of our knowledge, no other parameterized complexity results exist for the Roman Domination problem.

2.5 Our Results

The main contribution of the paper is the following:

- In Sect. 3.1 (resp. Sect. 3.2), we show that the RD-CVD (resp. IRD-CVD) problem is FPT.
- In Sect. 4, we show that the RD-CVD problem cannot be solved in time $2^{\epsilon k} n^{O(1)}$ (where $0 < \epsilon < 1$) unless SETH (refer to Conjecture 1) fails, neither it admits a polynomial kernel unless $\text{NP} \subseteq \text{coNP/poly}$.
- In Sect. 5, we conclude the paper with some future research directions.

3 Variants of Roman Domination Parameterized by CVD Size

In this section, we assume that a cluster vertex deletion set S of size k is given with the input graph $G = (V, E)$. If not, the algorithm mentioned in [2] can be used, which runs in $1.92^k n^{O(1)}$ time and outputs a cluster vertex deletion set of size at most k or concludes that there does not any cluster vertex deletion set of size at most k .

3.1 Roman Domination

In this section, we propose an FPT algorithm for the Roman Domination problem when the parameter is CVD size.

We consider a CVD set S as a part of the input, where $|S| = k$. Our algorithm starts with making a guess for $S_1 = V_1 \cap S$ and $S_2 = V_2 \cap S$, where (V_1, V_2) is an optimal Roman Dominating pair. At first, a guess of S_2 is made from S . Then,

the vertices of $N[S_2] \cap S$ are deleted from the graph. Then we guess S_1 from the remaining S and delete S_1 from S .

Note that S is a CVD set, hence $G \setminus S$ is disjoint union of cliques. Let $G \setminus S = C_1 \cup C_2 \cup \dots \cup C_q$, where every C_i is a clique and $|C_i| = \ell_i$, for $i \in [q]$. Note that $q \leq n - k$. After the selection of S_1 and S_2 , every clique belongs to exactly one of the following three types:

Type 0 (T_0) cliques: C_i is a T_0 clique if every vertex of C_i is adjacent to at least one vertex of S_2 .

Type 1 (T_1) cliques: C_i is a T_1 clique if exactly one vertex of C_i is not adjacent to any vertex of S_2 .

Type 2 (T_2) cliques: C_i is a T_2 clique if C_i contains at least two vertices which are not adjacent to any vertex of S_2 .

We define an order ρ_i on the vertices of the clique C_i as follows: if C_i is a T_0 or T_2 clique, then we order them arbitrarily; if C_i is a T_1 clique, then the set $C_i \setminus N[S_2]$ contains exactly one vertex. We make that vertex the first vertex of ρ_i and order the rest of the vertices of C_i arbitrarily. Now we define an order ρ on the vertex set of $G \setminus S$ as follows: $\rho = v_1, v_2, \dots, v_{|G \setminus S|}$, where first ℓ_1 vertices of ρ are vertices of C_1 and follows the order ρ_1 , then the next ℓ_2 vertices of ρ are vertices of C_2 and follows the order ρ_2 and so on. Now, here comes an observation.

Observation 1. *Given a T_2 clique C_i , any Roman Dominating pair (V_1, V_2) extended from (S_1, S_2) has one of the following properties:*

1. $C_i \cap V_2 \neq \emptyset$.
2. A Roman Dominating pair (V'_1, V'_2) can be extended from (S_1, S_2) , which has same or less weight than (V_1, V_2) and $V'_2 \cap C_i \neq \emptyset$

The proof of this observation is omitted due to space constraints. From the above observation, we can rephrase the remaining problem as follows:

RD-DISJOINTCLUSTER problem

Input: A graph $G = (V, E)$, a subset $S \subseteq V$ such that every component of $G \setminus S$ is a clique, a $(0, 1, 2)$ -flag vector $f = (f_1, f_2, \dots, f_q)$ corresponding to the cliques $(C_1, C_2, \dots, C_q)$ and $\ell \in \mathbb{Z}^+$.

Parameter: $|S|$.

Question: Does there exists a subset $T \subseteq G \setminus S$ which satisfies all of the following conditions?

- (a) For every C_i with $f_i = 2$, $T \cap C_i \neq \emptyset$.
- (b) $2|T| + g(T) \leq \ell$, where $g(T)$ = number of cliques with flag 1, which have empty intersection with T .

For an instance (G, S, ℓ) of the RD-CVD problem, with cliques $C_1, \dots, C_q$ and the guesses of S_1, S_2 , we build an instance $(\hat{G}, \hat{S}, f, \hat{\ell})$ of the RD-DISJOINTCLUSTER problem as follows:

- $\hat{G} = G \setminus ((N[S_2] \cap S) \cup S_1)$.

- $\hat{S} = S \setminus ((N[S_2] \cap S) \cup S_1)$.
- $\hat{\ell} = \ell - 2|S_2| - |S_1|$.
- For all $i \in [q]$, $f_i = j$, if C_i is a T_j clique, $j \in \{0, 1, 2\}$.

It is not hard to observe that the instance (G, S, ℓ) of the RD-CVD problem and the instance $(\hat{G}, \hat{S}, f, \hat{\ell})$ of the RD-DISJOINTCLUSTER problem are equivalent instances. This means (G, S, ℓ) is a YES instance if and only if $(\hat{G}, \hat{S}, f, \hat{\ell})$ is a YES instance.

Formulation of the Problem as a Variant of Set Cover: We define a variant of the set cover problem. Given an instance of the RD-DISJOINTCLUSTER problem, we construct an instance of the set cover problem. Let (G, S, f, ℓ) be an instance of the RD-DISJOINTCLUSTER problem and $\rho = (v_1, v_2, \dots, v_{|G \setminus S|})$ be an ordering of the vertex set of $G \setminus S$, as defined earlier. We take the universe $U = S$ and $F = \{S_1, S_2, \dots, S_{|G \setminus S|}\}$, where $S_i = N(v_i) \cap S$ for every $i \in [|G \setminus S|]$. Now, we modify the usual SET-COVER problem to suit our problem. The modified SET-COVER problem is defined below:

SET-COVERWITHPARTITION problem (SCP)

Input: Universe U , a family of sets $F = \{S_1, S_2, \dots, S_m\}$, a partition of $\beta = (\beta_1, \dots, \beta_q)$ of F , a $(0, 1, 2)$ flag vector $f = (f_1, f_2, \dots, f_q)$ corresponding to each block in the partition β and a non-negative integer ℓ .

Parameter: $|U|$.

Question: Does there exists a subset $F' \subseteq F$ which satisfies all of the following conditions?

- (a) For every β_i with $f_i = 2$, $F' \cap \beta_i \neq \emptyset$.
- (b) $2|F'| + g(F') \leq \ell$, where $g(A) =$ number of blocks with flag 1, which have empty intersection with A , for $A \subseteq F$.

Given an instance (G, S, f, ℓ) of the RD-DISJOINTCLUSTER problem, we define an instance (U, F, β, f', ℓ') of the SCP problem as follows:

- $U = S$.
- $F = \{S_i = N(v_i) \cap S \mid v_i \in G \setminus S\}$.
- $\beta_i = \{S_j \mid v_j \in C_i\}$, for $i \in [q]$.
- $f' = f$.
- $\ell' = \ell$.

It is not hard to show that the RD-DISJOINTCLUSTER and SCP are equivalent problems.

Observation 2. (G, S, f, ℓ) is a YES instance of the RD-DISJOINTCLUSTER problem if and only if (U, F, β, f', ℓ') is a YES instance of the SCP problem.

Now, we propose an algorithm to solve the SCP problem.

Theorem 2. *The SET-COVERWITHPARTITION problem can be solved in $2^{|U|}O(m \cdot |U|)$ time.*

Proof. We propose a dynamic programming algorithm to solve the problem. For every $W \subseteq U$, $j \in [m]$ and $b \in \{0, 1, 2\}$, we define $OPT[W, j, b] := \min_X \{2|X| + g_j(X)\}$, where X satisfies the following properties:

1. $X \subseteq \{S_1, \dots, S_j\}$.
2. X covers W .
3. Let β_x be the block that contains S_j . We redefine $f_x = b$, where f_x is the flag associated with β_x . From every block β_i ($i \leq x$) with $f_i = 2$, at least one set from β_i is in X .
4. The function g_j is defined as follows. $g_j(X) :=$ number of blocks β_i ($i \leq x$) with $f_i = 1$, which have empty intersection with X .

Now, coming to the base case, for every $W \subseteq U$, with $W \neq \emptyset$ and $b \in \{0, 1, 2\}$; $OPT[W, 1, b] = 2$ if $W \subseteq S_1$, $OPT[W, 1, b] = \infty$, otherwise.

If $W = \emptyset$, $OPT[W, 1, b] = b$, for $b \in \{0, 1, 2\}$. To compute all the values of $OPT[W, j, b]$, we initially set all the remaining values to be ∞ . We construct the following recursive formulation for $OPT[W, j+1, b]$, for $j \geq 1$:

Case 1: S_{j+1} is not the first set of the block β_x .

Note that two possibilities appear here. First, we pick S_{j+1} in the solution X . Hence, we are left with the problem of covering $W \setminus S_{j+1}$ with some subset of $\{S_1, \dots, S_j\}$ and since S_{j+1} from the partition β_x is already taken in solution, so the flag of β_x can be reset to 0. Hence, in this case $OPT[W, j+1, b] = 2 + OPT[W \setminus S_{j+1}, j, 0]$.

In the latter case, we do not pick S_{j+1} in X ; hence nothing is changed except the fact that now we need to cover W with a subset of $\{S_1, \dots, S_j\}$ and the flag of β_x remains unchanged as b . Hence, $OPT[W, j+1, b] = OPT[W, j, b]$.

So, $OPT[W, j+1, b] = \min\{2 + OPT[W \setminus S_{j+1}, j, 0], OPT[W, j, b]\}$.

Case 2: S_{j+1} is the first set of the block β_x . Here, three scenarios can appear:

Case 2.1: $b = 2$.

In this case, there is no option but to include S_{j+1} in the solution as $b = 2$. Hence, we take S_{j+1} in the solution and shift to the previous block. Now we need to cover $W \setminus S_{j+1}$ with a subset of $\{S_1, \dots, S_j\}$. Hence, $OPT[W, j+1, b] = 2 + OPT[W \setminus S_{j+1}, j, f_{x-1}]$.

Case 2.2: $b = 0$.

In this case, there are two choices, to include S_{j+1} in the solution or not. If we include S_{j+1} in the solution, then $OPT[W, j+1, b] = 2 + OPT[W \setminus S_{j+1}, j, f_{x-1}]$. If we do not, then $OPT[W, j+1, b] = OPT[W, j, f_{x-1}]$. Hence $OPT[W, j+1, b] = \min\{2 + OPT[W \setminus S_{j+1}, j, f_{x-1}], OPT[W, j, f_{x-1}]\}$

Case 2.3: $b = 1$.

Similarly, in this case, there are two choices. If S_{j+1} is included in the solution then $OPT[W, j+1, b] = 2 + OPT[W \setminus S_{j+1}, j, f_{x-1}]$, by similar argument as above. If not, then S_{j+1} has to contribute 1 in $OPT[W, j+1, b]$, as at least one set from the block β_x has to contribute 1 to $OPT[W, j+1, b]$ and S_{j+1} is the only set left

in β_x at this moment. So, in this case, $OPT[W, j+1, b] = 1 + OPT[W, j, f_{x-1}]$. Hence, $OPT[W, j+1, b] = \min\{2 + OPT[W \setminus S_{j+1}, j, f_{x-1}], 1 + OPT[W, j, f_{x-1}]\}$.

We compute $OPT[W, j, b]$ in the increasing order of size of W, j, b . Hence, there are $3 \cdot 2^{|U|} \cdot m$ subproblems. It takes $|U|$ time to compute set differences (like $W \setminus S_{j+1}$). Hence, the time-complexity of our algorithm is $2^{|U|}O(m \cdot |U|)$. $\square$

Hence, the following corollary can be concluded.

Corollary 1. *The RD-DISJOINTCLUSTER problem can be solved in time $2^{|S|}n^{O(1)}$.*

Theorem 3. *The RD-CVD problem can be solved in time $4^k n^{O(1)}$.*

Proof. Given an instance (G, S, ℓ) of the RD-CVD problem and for every guess of $S'_1, S'_2 \subseteq S$ (with $|S'_1| = i_1$ and $|S'_2| = i_2$), we can construct an instance $(\hat{G}, \hat{S}, f, \hat{\ell})$ of the RD-DISJOINTCLUSTER problem, which can be solved in time $2^{k-i_1-i_2}n^{O(1)}$. Hence, total time taken is $\sum_{i_1=1}^k \binom{k}{i_1} \sum_{i_2=1}^{k-i_1} \binom{k-i_1}{i_2} 2^{k-i_1-i_2} n^{O(1)} = 4^k n^{O(1)}$. $\square$

In the next section, using a similar approach, we show that the IRD-CVD problem is also FPT.

3.2 Independent Roman Domination

In this section, we propose an FPT algorithm for the Independent Roman Domination problem when the parameter is the CVD size.

Similarly, like the case of Roman Domination, a guess for $S_1 = V_1 \cap S$ and $S_2 = V_2 \cap S$ is made, where (V_1, V_2) is an optimal independent Roman Dominating pair. At first, we guess an independent set S_2 from S and then delete all the vertices of $N[S_2]$ from the graph, as if our choice of S_2 is right, then all the vertices in $N(S_2) \setminus S_2$ should have label 0. Then, we choose another independent set S_1 from the remaining S and delete all the vertices of S_1 and $N[S_1] \cap (G \setminus S)$ from the remaining graph. Note that if there exists a clique $C_i \subseteq N[S_1] \cup N[S_2]$ such that C_i has a vertex v , that is not adjacent to any vertex of S_2 , but it is adjacent to some vertex in S_1 , then our choices of S_1, S_2 are incorrect, and we do not move further with these choices of S_1 and S_2 .

Note that S is a CVD set, hence $G \setminus S$ is disjoint union of cliques. Let $G \setminus S = C_1 \cup C_2 \cup \dots \cup C_q$, where every C_i is a clique and $|C_i| = \ell_i$, for $i \in [q]$. Note that $q \leq n - k$. Note that, after the selection of S_1 and S_2 and the deletion process, every clique belongs to exactly one of the following two types:

Type 1 (T_1) cliques: C_i is a T_1 clique if C_i has exactly one vertex.

Type 2 (T_2) cliques: C_i is a T_2 clique if C_i has at least two vertices.

Observation 3. *Given a T_2 clique C_i , any independent Roman Dominating pair (V_1, V_2) extended from (S_1, S_2) has the following property: $C_i \cap V_2 \neq \emptyset$.*

Proof. C_i is a T_2 clique, hence there exist at least two vertices v_1, v_2 in C_i . Note that both of them can not have non zero labels; at least one of them must have label 0. Without loss of generality, let v_1 have label 0, but v_1 does not have any neighbor in S , which has label 2, which implies a vertex in C_i must have label 2. Hence, the result follows. $\square$

Hence the remaining problem can be rephrased as follows:

IRD-DISJOINTCLUSTER problem

Input: A graph $G = (V, E)$, a subset $S \subseteq V$ such that every component of $G \setminus S$ is a clique, a $(1, 2)$ -flag vector $f = (f_1, f_2, \dots, f_q)$ corresponding to the cliques $(C_1, C_2, \dots, C_q)$ and $\ell \in \mathbb{Z}^+$.

Parameter: $|S|$.

Question: Does there exists a subset $T \subseteq G \setminus S$ which satisfies all of the following conditions?

- (a) For every C_i with $f_i = 2$, $|T \cap C_i| = 1$.
- (b) $2|T| + g(T) \leq \ell$, where $g(T)$ = number of cliques with flag 1, which have empty intersection with T .

For an instance (G, S, ℓ) of the IRD-CVD problem, with cliques $C_1, \dots, C_q$ and the guesses of S_1, S_2 , we build an instance $(\hat{G}, \hat{S}, f, \hat{\ell})$ of the IRD-DISJOINTCLUSTER problem as follows:

- $\hat{G} = G \setminus (S_1 \cup N[S_2] \cup (N(S_1) \cap (G \setminus S)))$.
- $\hat{S} = S \setminus (N[S_2] \cup S_1)$.
- $\hat{\ell} = \ell - 2|S_2| - |S_1|$.
- $f_i = j$, if C_i is a T_j clique, $i \in [q]$ and $j \in \{1, 2\}$.

Formulation of the problem as a variant of set cover: We define a variant of the set cover problem similarly to the Roman Domination problem. Given an instance of the IRD-DISJOINTCLUSTER problem, we construct an instance of the set cover problem. Let (G, S, ℓ, f) be an instance of the IRD-DISJOINTCLUSTER problem and ρ be any arbitrary order of the vertex set $G \setminus S$. We take the universe $U = S$ and $F = \{S_1, S_2, \dots, S_{|G \setminus S|}\}$, where $S_i = N(v_i) \cap S$ for every $i \in [|G \setminus S|]$. Now, we modify the usual set cover problem to suit our problem. The modified set cover problem is defined below:

INDEPENDENT-SET-COVERWITHPARTITION problem (ISCP)

Input: Universe U , a family of sets $F = \{S_1, S_2, \dots, S_m\}$, a partition of $\beta = (\beta_1, \dots, \beta_q)$ of F , a $(1, 2)$ flag vector $f = (f_1, f_2, \dots, f_q)$ corresponding to each block in the partition β and a non-negative integer ℓ .

Parameter: $|U|$.

Question: Does there exists a subset $F' \subseteq F$ which satisfies all of the following conditions?

- (a) For every β_i with $f_i = 2$, $|F' \cap \beta_i| = 1$.
- (b) $2|F'| + g(F') \leq \ell$, where $g(A)$ = number of blocks with flag 1, which have empty intersection with A ; for $A \subseteq F$.

Given an instance (G, S, f, ℓ) of the IRD-DISJOINTCLUSTER problem, we define an instance (U, F, β, f', ℓ') of the ISCP problem as follows:

- $U = S$.
- $F = \{S_i = N(v_i) \cap S \mid v_i \in G \setminus S\}$.
- $\beta_i = \{S_j \mid v_j \in C_i\}$, for $i \in [q]$.
- $f' = f$.
- $\ell' = \ell$.

Observation 4. (G, S, f, ℓ) is a YES instance of the IRD-DISJOINTCLUSTER problem if and only if (U, F, β, f', ℓ') is a YES instance of the ISCP problem.

Next, we propose an algorithm to solve the ISCP problem.

Theorem 4. The INDEPENDENT-SET-COVERWITHPARTITION problem can be solved in $2^{|U|}O(m \cdot |U|)$ time.

Proof. The proof is omitted due to space constraints. □

Hence, the following corollary can be concluded.

Corollary 2. The IRD-DISJOINTCLUSTER problem can be solved in time $2^{|S|}n^{O(1)}$.

Theorem 5. The IRD-CVD problem can be solved in time $4^k n^{O(1)}$.

Proof. The proof is analogous to the proof of Theorem 3. □

4 Lower Bounds

In this section, we propose a lower bound on the time-complexity of the RD-CVD problem. We also show that the RD-CVD (resp. RD-VC) problem does not admit a polynomial kernel (recall that RD-VC is the Roman Domination problem parameterized by vertex cover number).

First, we provide the lower bound for the RD-CVD problem. Below, we state a necessary result from the existing literature (refer to Theorem 1.1 in [3]).

Theorem 6. [3] The following statement is equivalent to SETH: For every $\epsilon < 1$, there exists $d \in \mathbb{Z}^+$, such that the d -HITTING SET problem for set systems over $[n]$ can not be solved in time $O(2^{\epsilon n})$.

Now, we show a reduction from the d -HITTING SET problem to the RD-CVD problem to show a similar lower bound like Theorem 6 for the RD-CVD problem.

Theorem 7. There exists a polynomial time algorithm that takes an instance (U, F, t) of the d -HITTING SET problem and outputs an instance $(G, 2t)$ of the RD-CVD problem (and the RD-VC problem), where G has a cluster vertex deletion set (and vertex cover) of size $|U|$; and (U, F) has a d -HITTING SET of size at most t if and only if G has a Roman Dominating function of size at most $2t$.

Proof. The proof is omitted due to space constraints. $\square$

Hence, by Theorem 6 and 7, we can prove the following theorem.

Theorem 8. *The RD-CVD (and RD-VC) problem can not be solved in time $2^{\epsilon k} n^{O(1)}$, for any $0 < \epsilon < 1$, unless SETH fails.*

Proof. Let the RD-CVD (or RD-VC) problem be solved in time $2^{\epsilon k} n^{O(1)}$. Then by Theorem 7, we can solve the d -HITTING SET problem with $|U| = k$ in $2^{\epsilon k} n^{O(1)}$ time. Hence, by Theorem 6, this contradicts the SETH. Hence, the RD-CVD (and RD-VC) problem can not be solved in time $2^{\epsilon k} n^{O(1)}$, unless the SETH fails. $\square$

We state another theorem to show that the RD-CVD (and RD-VC) problem is unlikely to admit a polynomial kernel.

Theorem 9. [4] *The d -HITTING SET problem parameterized by the universe size does not admit any polynomial kernel unless $NP \subseteq coNP/poly$.*

Hence combining the above discussion with Theorem 9, the following theorem can be concluded.

Theorem 10. *The RD-CVD (and RD-VC) problem does not admit a polynomial kernel unless $NP \subseteq coNP/poly$.*

Proof. By Theorem 9, the d -HITTING SET problem parameterized by universe size does not admit a polynomial kernel unless $NP \subseteq coNP/poly$. Since the reduction provided in Theorem 7 is a polynomial parameter transformation (PPT), the RD-CVD and RD-VC do not admit a polynomial kernel unless $NP \subseteq coNP/poly$. $\square$

5 Conclusion

In this work, we have extended the study on the parameterized complexity of the Roman Domination problem and one of its variants. There are other interesting structural parameters, such as neighborhood diversity and cliquewidth, for which it would be interesting to determine whether the problem parameterized by these parameters is fixed parameter tractable (FPT).

Another promising research direction is to develop an algorithm to solve the RD-CVD problem with better time-complexity than $4^k n^{O(1)}$. Given that the lower bound on time-complexity mentioned in Theorem 8, it might be possible to achieve an improved algorithm.

Acknowledgement. Research of Arti Pandey is supported by CRG project, Grant Number-CRG/2022/008333, Science and Engineering Research Board (SERB), India.

References

1. Alber, J., Fellows, M.R., Niedermeier, R.: Polynomial-time data reduction for dominating set. *J. ACM* **51**(3), 363–384 (2004)
2. Boral, A., Cygan, M., Kociumaka, T., Pilipczuk, M.: A fast branching algorithm for cluster vertex deletion. *Theory Comput. Syst.* **58**(2), 357–376 (2016)
3. Cygan, M., et al.: On problems as hard as CNF-SAT. *ACM Trans. Algorithms* (2016)
4. Dom, M., Lokshtanov, D., Saurabh, S.: Kernelization lower bounds through colors and ids. *ACM Trans. Algorithms* **11** (2014)
5. Downey, R.G., Fellows, M.R.: Fixed-parameter tractability and completeness II: on completeness for W. *Theor. Comput. Sci.* **141**(1&2), 109–131 (1995)
6. Fernau, H.: Roman domination: a parameterized perspective. *Int. J. Comput. Math.* **85**(1), 25–38 (2008)
7. Fomin, F.V., Kratsch, D.: *Exact Exponential Algorithms*. Texts in Theoretical Computer Science. An EATCS Series. Springer, Cham (2010)
8. Goyal, D., Jacob, A., Kumar, K., Majumdar, D., Raman, V.: Parameterized complexity of dominating set variants in almost cluster and split graphs. *CoRR*, abs/2405.10556 (2024)
9. Impagliazzo, R., Paturi, R., Zane, F.: Which problems have strongly exponential complexity? *J. Comput. Syst. Sci.* **63**(4), 512–530 (2001)
10. Mohanapriya, A., Renjith, P., Sadagopan, N.: Roman k-domination: hardness, approximation and parameterized results. In Lin, C.-C., Lin, B.M.T., Liotta, G. (eds.) *WALCOM: Algorithms and Computation - 17th International Conference and Workshops, WALCOM 2023, Hsinchu, Taiwan, 22–24 March 2023, Proceedings*. Lecture Notes in Computer Science, vol. 13973, pp. 343–355. Springer, Cham (2023)
11. Philip, G., Raman, V., Sikdar, S.: Polynomial kernels for dominating set in graphs of bounded degeneracy and beyond. *ACM Trans. Algorithms* **9**(1), 11:1–11:23 (2012)
12. Stewart, I.: Defend the roman empire! *Sci. Am.* **281**, 136–138 (1999)